**Question:**

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

**Reasoning + Answer:**

<think> To solve this problem, let's define a variable to help us work through it systematically. Let's say Melanie started with \( x \) vacuum cleaners. According to the problem, she sold a third of her vacuum cleaners at the green house. So, the number of vacuum cleaners sold at the green house is \( \frac{x}{3} \).

After selling at the green house, the number of vacuum cleaners left with Melanie would be \( x - \frac{x}{3} \). Simplifying this, we get \( x - \frac{x}{3} = \frac{3x}{3} - \frac{x}{3} = \frac{2x}{3} \).

Next, the problem states that she sold 2 more vacuum cleaners at the red house. So, we need to subtract these 2 cleaners from the remaining vacuum cleaners after the green house sale. Thus, after the red house sale, the number of vacuum cleaners left would be \( \frac{2x}{3} - 2 \).

According to the problem, she then sold half of what was left at the orange house. Let's denote the remaining vacuum cleaners after the red house sale as \( \frac{2x}{3} - 2 \). Since she sold half of this amount at the orange house, the number of vacuum cleaners sold at the orange house is \( \frac{1}{2} \left( \frac{2x}{3} - 2 \right) \). After selling at the orange house, the number of vacuum cleaners left with her is \( \frac{2x}{3} - 2 - \frac{1}{2} \left( \frac{2x}{3} - 2 \right) \). We know that after these sales, she has 5 vacuum cleaners left. So, let's set up the equation using this information:

\[ \frac{2x}{3} - 2 - \frac{1}{2} \left( \frac{2x}{3} - 2 \right) = 5 \]

To simplify and solve for \( x \), we first distribute and combine like terms. Let's start by simplifying the right side of the equation:

\[ \frac{2x}{3} - 2 - \frac{1}{2} \left( \frac{2x}{3} - 2 \right) = 5 \]
\[ \frac{2x}{3} - 2 - \left( \frac{2x}{6} - 1 \right) = 5 \] // Distribute \(-\frac{1}{2}\) into \(\frac{2x}{3} - 2\)
\[ \frac{2x}{3} - 2 - \frac{2x}{6} + 1 = 5 \] // Simplify \(-2 - 1\) to \(-3\)

Now, to combine \(\frac{2x}{3}\) and \(-\frac{2x}{6}\), we note that \(\frac{2x}{3} = \frac{4x}{6}\):

\[ \frac{4x}{6} - \frac{2x}{6} - 3 + 1 = 5 \]
\[ \frac{2x}{6} - 3 + 1 = 5 \] // Combine like terms \(\frac{2x}{6}\) and \(-3 + 1\) to \(\frac{x}{3} - 2\)
\[ \frac{x}{3} - 2 = 5 \]

Next, we'll solve for \( x \) by isolating it on one side of the equation. Add 2 to both sides:

\[ \frac{x}{3} = 7 \]

Now, multiply both sides by 3 to solve for \( x \):

\[ x = 7 \times